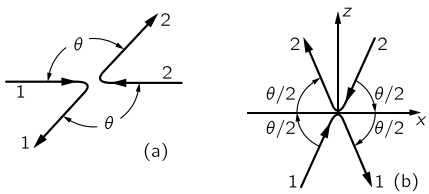

Figure 4-2 Two views of an elastic collision between equal objects moving at the same speed in opposite directions.

Then how does the collision look? It looks as though particle 1 is just going straight up, because it has lost its horizontal component, and it comes straight down again, also because it does not have that component. That is, the collision appears as shown in Figure 4-3(a). Particle 2, however, was going the other way, and as we ride past it appears to fly by at some terrific speed and at a smaller angle, but we can appreciate that the angles before and after the collision are the *same*. Let us denote by u the horizontal component of the velocity of particle 2, and by w the vertical velocity of particle 1.

Now the question is, what is the vertical velocity $u \tan \alpha$? If we knew that, we could get the correct expression for the momentum, using the law of conservation of momentum in the vertical direction. Clearly, the horizontal component of the momentum is conserved: it is the same before and after the collision for both particles, and is zero for particle 1. So we need use the conservation law only for the upward velocity $u \tan \alpha$. *But we can* get the upward velocity, simply by looking at the same collision going the other way! If we look at the collision of Figure 4-3(a) from a car moving to the left with speed u, we see the same collision, except "turned over," as shown in Figure 4-3(b). Now particle 2 is the one that goes up and down with speed w, and particle 1 has picked up the horizontal speed u. Of course, now we *know* what the velocity $u \tan \alpha$ is: it is $w\sqrt{1 - u^2/c^2}$ (see Eq. 4.7). We know that the change in the vertical momentum of the vertically moving particle is

$$\Delta p = 2m_w w$$